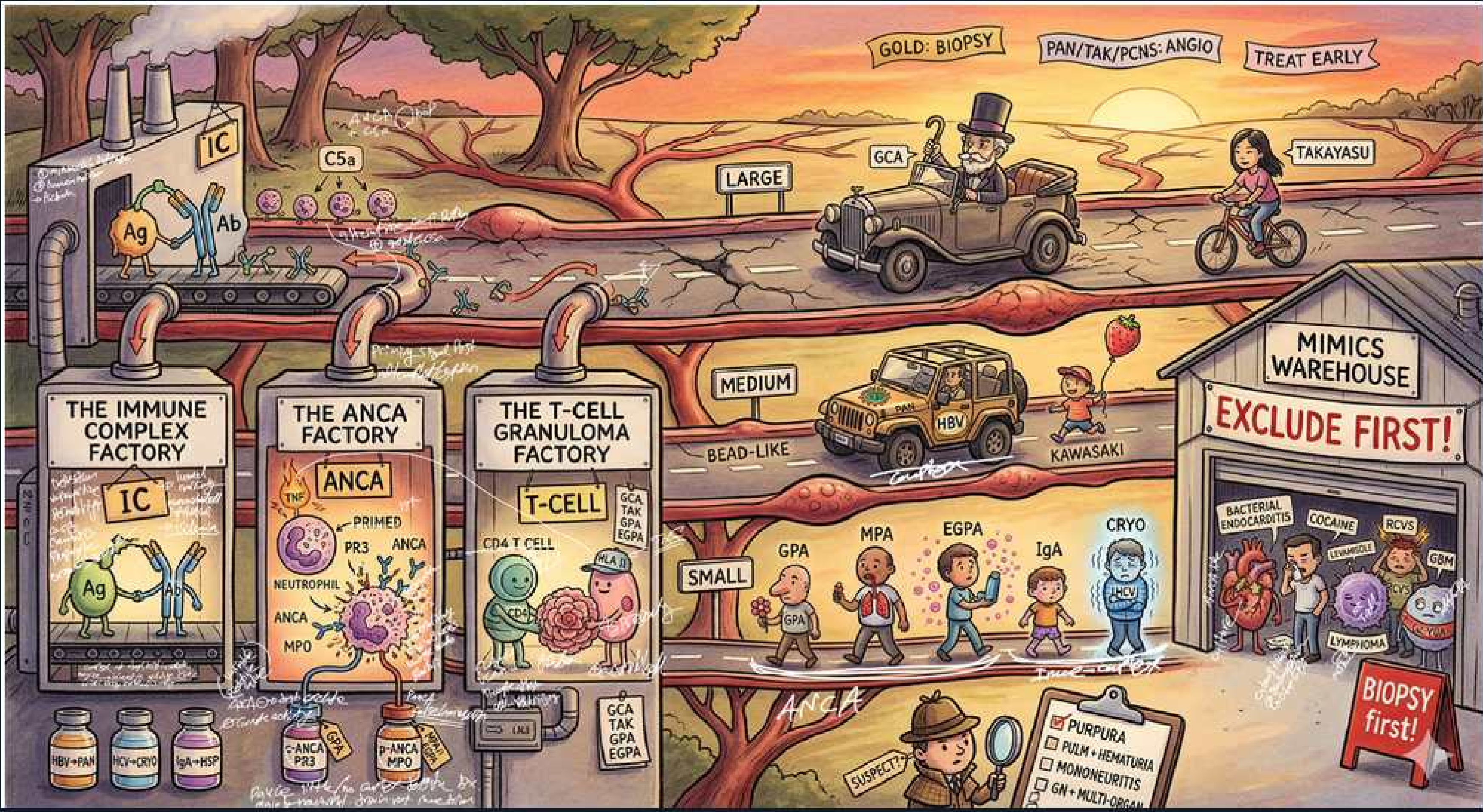
# Vasculitis — Classification & Pathology



## 1. Immune Complex Factory

**WHAT YOU SEE**

Factory labeled THE IMMUNE COMPLEX FACTORY with Ag-Ab (IC) deposits

**WHAT IT ENCODES**

Immune-complex small-vessel vasculitis is a type III hypersensitivity reaction: circulating antigen-antibody complexes deposit in vessel walls, fix complement, and recruit neutrophils. Serum complement (C3/C4) is therefore consumed and LOW. This group includes IgA vasculitis (HSP), cryoglobulinemic vasculitis, hypocomplementemic urticarial vasculitis, and anti-GBM disease. Immunofluorescence shows granular ('lumpy-bumpy') immune deposits — the opposite of pauci-immune AAV.

**BOARD TRAP**

Immune-complex vasculitides are ANCA-NEGATIVE with LOW complement and granular IF. Lumping them with ANCA-associated vasculitis (which is ANCA-positive, normal complement, pauci-immune) is the trap. The complement level and IF pattern are the discriminators: low C3/C4 + granular = immune-complex; normal C3/C4 + pauci-immune = AAV.

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## 2. C5a / complement

**WHAT YOU SEE**

C5a tag near the immune complex machinery

**WHAT IT ENCODES**

Complement activation generates C5a, a potent anaphylatoxin/chemoattractant that primes and recruits neutrophils, amplifying vessel-wall inflammation. In immune-complex disease this consumes serum complement (low C3/C4). The C5a axis is also a therapeutic target in ANCA vasculitis — avacopan is an oral C5a-receptor antagonist used as a steroid-sparing agent.

**BOARD TRAP**

Don't assume complement is consumed in every vasculitis. In ANCA-associated vasculitis serum complement is typically NORMAL despite the alternative-pathway C5a contribution to pathology; low complement instead points to immune-complex disease (cryoglobulinemia, lupus, post-strep).

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## 3. ANCA Factory

**WHAT YOU SEE**

Factory labeled THE ANCA FACTORY with PR3, MPO, primed neutrophil

**WHAT IT ENCODES**

ANCA-associated vasculitis (GPA, MPA, EGPA) is pauci-immune: cytokine-primed neutrophils express PR3/MPO on their surface, ANCA antibodies bind and trigger a respiratory burst with degranulation, directly injuring small vessels (necrotizing capillaritis). Because injury is antibody/neutrophil-mediated rather than complex-mediated, there are few/no immune deposits on IF. Serum complement is usually normal.

**BOARD TRAP**

'Pauci-immune' means scant/absent immune-complex deposition — the diagnostic opposite of immune-complex vasculitis. Expecting heavy granular deposits or low complement in AAV is wrong; their absence is what defines AAV histologically.

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## 4. PR3 vs MPO

**WHAT YOU SEE**

Labels PR3 (c-ANCA) and MPO (p-ANCA) inside ANCA factory

**WHAT IT ENCODES**

PR3-ANCA produces a cytoplasmic (c-ANCA) immunofluorescence pattern and predominates in GPA. MPO-ANCA produces a perinuclear (p-ANCA) pattern and predominates in MPA and EGPA. Serotype increasingly guides prognosis better than clinical label: PR3-disease relapses more and responds especially well to rituximab; MPO-disease relapses less but carries more chronic renal damage. Always confirm IIF patterns with antigen-specific PR3/MPO ELISA.

**BOARD TRAP**

p-ANCA is not specific for vasculitis — it can be positive in IBD, autoimmune hepatitis, and drug effects (often as atypical/x-ANCA without MPO). A positive p-ANCA without anti-MPO on ELISA in a patient with colitis is not vasculitis. Confirm with the antigen-specific assay before diagnosing AAV.

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## 5. T-Cell Granuloma Factory

**WHAT YOU SEE**

Factory labeled THE T-CELL GRANULOMA FACTORY with CD4 T-cell

**WHAT IT ENCODES**

Granulomatous vasculitis is driven by CD4 T-cell/macrophage responses forming granulomas in vessel walls. Large-vessel granulomatous disease (GCA, Takayasu) features giant cells and disrupted internal elastic lamina; GPA and EGPA add necrotizing granulomas in small/medium vessels. This Th1/Th17 mechanism is why IL-6 blockade (tocilizumab) works in GCA.

**BOARD TRAP**

Granulomatous vasculitis spans vessel sizes (large: GCA/Takayasu; small: GPA/EGPA), so 'granuloma' alone doesn't pin the diagnosis. PAN and MPA are NON-granulomatous — selecting PAN/MPA when biopsy shows granulomas is the trap.

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## 6. Large vessel road / car

**WHAT YOU SEE**

Old man in vintage car labeled GCA on a road tagged LARGE

**WHAT IT ENCODES**

Large-vessel vasculitis = giant cell (temporal) arteritis and Takayasu arteritis; both involve the aorta and its major branches. GCA: age >50, headache, jaw claudication, vision loss, PMR overlap, very high ESR, treated with urgent steroids. Both raise the long-term risk of aortic aneurysm/dissection, warranting imaging surveillance.

**BOARD TRAP**

Large-vessel vasculitis is diagnosed largely by IMAGING (temporal artery US/biopsy for GCA; CTA/MRA/PET for both), not by renal/skin biopsy. Expecting palpable purpura or glomerulonephritis (small-vessel features) in GCA/Takayasu is the trap — large-vessel disease causes ischemia/claudication, not purpura.

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## 7. Takayasu cyclist

**WHAT YOU SEE**

Young woman on a bicycle labeled TAKAYASU

**WHAT IT ENCODES**

Takayasu arteritis is large-vessel granulomatous aortitis of YOUNG patients (typically <40-50), with a strong female and Asian predominance. 'Pulseless disease': arm blood-pressure discrepancies, diminished pulses, bruits, and limb claudication from aortic-arch branch stenoses. Diagnosed by CTA/MRA/PET; treated with glucocorticoids ± steroid-sparing agents.

**BOARD TRAP**

Age is the key discriminator from GCA: under ~40-50 think Takayasu; over 50 think GCA. Diagnosing GCA in a 28-year-old woman with subclavian stenosis is wrong — GCA essentially never occurs under 50.

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## 8. Medium vessel jeep

**WHAT YOU SEE**

Jeep labeled HBV on a road tagged MEDIUM, bead-like aneurysms

**WHAT IT ENCODES**

Medium-vessel vasculitis = polyarteritis nodosa (PAN) and Kawasaki disease. PAN: necrotizing arteritis of muscular arteries, often hepatitis-B-associated, causing renal microaneurysms ('beads on a string'), mononeuritis multiplex, GI ischemia, skin nodules/livedo, and renovascular HTN. PAN is ANCA-NEGATIVE and spares the lungs and glomeruli.

**BOARD TRAP**

Medium-vessel disease (PAN) gives microaneurysms/infarcts, NOT glomerulonephritis and NOT pulmonary capillaritis — those are small-vessel/ANCA features. A 'vasculitis with RPGN or alveolar hemorrhage' is not PAN; that's MPA/GPA. Also, PAN is ANCA-negative, so a positive ANCA argues against PAN.

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## 9. Kawasaki child

**WHAT YOU SEE**

Child with balloon labeled KAWASAKI beside medium-vessel road

**WHAT IT ENCODES**

Kawasaki disease is a medium-vessel vasculitis of children <5 yr: fever ≥5 days plus conjunctivitis (non-exudative), oral changes (strawberry tongue/cracked lips), polymorphous rash, extremity changes (edema/desquamation), and cervical lymphadenopathy. Untreated, ~25% develop coronary artery aneurysms. Treat early (within 10 days) with IVIG 2 g/kg + aspirin.

**BOARD TRAP**

The danger is delayed treatment: IVIG given after day 10 is far less protective against coronary aneurysms. Withholding IVIG to 'observe' a child meeting Kawasaki criteria is the trap. Incomplete Kawasaki (fewer criteria but high inflammatory markers + coronary changes) still warrants treatment.

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## 10. Small vessel kids row

**WHAT YOU SEE**

Row of figures GPA, MPA, EGPA, IgA, CRYO labeled SMALL

**WHAT IT ENCODES**

Small-vessel vasculitis splits into two mechanisms: (1) ANCA-associated/pauci-immune — GPA, MPA, EGPA (normal complement, ANCA+); and (2) immune-complex — IgA vasculitis (HSP), cryoglobulinemic, hypocomplementemic urticarial, anti-GBM (low complement, ANCA−, granular IF). Both can cause palpable purpura and glomerulonephritis, but the serologic/IF signatures separate them.

**BOARD TRAP**

Palpable purpura + glomerulonephritis is the shared small-vessel phenotype — the wrong move is stopping at 'small-vessel vasculitis.' You must subtype with ANCA, complement, cryoglobulins, and IF (pauci-immune vs granular) because induction therapy and treating the underlying cause (HCV, drug) differ.

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## 11. Gold: Biopsy banner

**WHAT YOU SEE**

Banner GOLD: BIOPSY across top

**WHAT IT ENCODES**

Tissue biopsy is the gold-standard confirmation for most vasculitides (skin, kidney, lung, nerve, temporal artery), establishing the histologic pattern — pauci-immune vs granular IF, granulomatous vs not, vessel size. Biopsy the most accessible involved organ. It also excludes mimics (infection, malignancy, calciphylaxis) before committing to immunosuppression.

**BOARD TRAP**

Biopsy is gold standard for MOST vasculitides, but key exceptions are diagnosed by imaging instead (PAN, Takayasu, primary CNS vasculitis — see next). Insisting on renal/visceral biopsy in PAN (risking bleeding from microaneurysms) is the trap; use angiography there.

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## 12. PAN/Tak/PCNS: Angio

**WHAT YOU SEE**

Banner PAN/TAK/PCNS: ANGIO

**WHAT IT ENCODES**

For three vasculitides angiography is preferred over biopsy: PAN (renal/mesenteric catheter angiography shows microaneurysms and beading), Takayasu (CTA/MRA/PET shows wall thickening and stenoses of the aorta/branches), and primary CNS vasculitis (where brain biopsy is high-risk and patchy, though it remains the true gold standard when feasible). Tissue in these vessels is hard or hazardous to obtain.

**BOARD TRAP**

Choosing renal biopsy for suspected PAN is the classic trap — it risks hemorrhage from microaneurysms and shows non-specific changes; mesenteric/renal angiography is the diagnostic test. For Takayasu, biopsy is rarely needed at all.

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## 13. Treat Early banner

**WHAT YOU SEE**

Banner TREAT EARLY top right

**WHAT IT ENCODES**

Early treatment prevents irreversible organ/vessel damage. The paradigm example is GCA: start high-dose glucocorticoids IMMEDIATELY on clinical suspicion, BEFORE temporal artery biopsy, to protect vision — the biopsy stays positive for ~1-2 weeks after steroids begin. Similarly, early IVIG protects coronaries in Kawasaki, and prompt induction limits renal loss in AAV.

**BOARD TRAP**

The trap is delaying steroids in GCA to await biopsy results — vision loss is often sudden and irreversible, and the contralateral eye is at risk. Steroids first do not 'ruin' the biopsy within the first 1-2 weeks.

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## 14. Mimics Warehouse — Exclude First

**WHAT YOU SEE**

Warehouse labeled MIMICS WAREHOUSE / EXCLUDE FIRST with bacterial, drugs, cocaine, malignancy

**WHAT IT ENCODES**

Before immunosuppressing, exclude vasculitis mimics: infective endocarditis (can cause purpura, GN, and even ANCA positivity), other infections (HCV/HBV, HIV), drugs (cocaine/levamisole, hydralazine, PTU, minocycline causing drug-induced ANCA), atrial myxoma/cholesterol emboli, calciphylaxis, and malignancy (lymphoma, intravascular lymphoma). Blood cultures, echocardiography, and a drug history are essential.

**BOARD TRAP**

Treating a mimic as vasculitis is dangerous: giving steroids/cyclophosphamide to a patient who actually has infective endocarditis (which can be ANCA-positive) can be lethal. The discriminator is to draw blood cultures and get an echo BEFORE immunosuppression in any ANCA-positive patient with fever and a new murmur.

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